

AWS

Explained Simply

The cheat sheet I wish I'd had
when I started.

40 must-know services. One sentence each, plus an everyday comparison. No jargon.



Compute



Storage



Databases



Networking & delivery



Security & identity



Monitoring & operations



Messaging & integration



AI & machine learning



Developer & deployment tools

by

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Swipe →

How to read this

The two ideas that make the rest click



AWS IN ONE SENTENCE

Amazon Web Services lets you rent computers, storage and ready-made tools over the internet, and pay only for what you use.

Region: e.g. London

eu-west-2



Zone A

own power



Zone B

own network



Zone C

miles apart

REGION

A city where AWS has data centres. Pick one near your users (and your data rules).

AVAILABILITY ZONE

A separate building in that city. Use 2+ so one outage won't take you down.

EACH CARD SHOWS

Service name → what it does in one line → an everyday comparison or a “use it when” example.

Before you experiment: set a spending alert in AWS Budgets, so a forgotten test server can't surprise you.

Compute

Where your code actually runs



Amazon EC2 VIRTUAL SERVERS

Rent virtual servers and control everything on them.

Think of it as: renting a flat: you pick the size, then furnish and maintain it yourself (OS, software, patches).

AWS Lambda SERVERLESS

Runs your code only when triggered. No servers to manage.

Think of it as: a taxi: pay per trip instead of owning a car. Great for short tasks (up to 15 minutes per run).

AWS Elastic Beanstalk EASY MODE

Upload your app; it sets up the servers and scaling for you.

Use it when: you want a standard web app online fast, without wiring up every piece yourself.

Amazon ECS & AWS Fargate CONTAINERS

ECS runs your containers. Fargate removes the servers.

Think of it as: shipping containers: pack your app once, run it anywhere. With Fargate, AWS drives the lorry.

Amazon EKS KUBERNETES

Managed Kubernetes: AWS runs the control plane for you.

Use it when: your team already knows Kubernetes, or needs the same setup across clouds and on-premises.

Storage

Where your files and data live



Amazon S3 OBJECT STORAGE

Stores any amount of files in 'buckets', reachable via the web.

Think of it as: a bottomless, very reliable Dropbox for apps: images, backups, logs, even whole static websites.

Amazon EBS BLOCK STORAGE

A virtual hard drive attached to your EC2 server.

Think of it as: an external SSD plugged into your server: fast and movable, but it stays in one Availability Zone.

Amazon EFS SHARED FILES

A shared file system that many servers can use at once.

Think of it as: the shared office drive everyone opens at the same time (for Linux servers). It grows and shrinks by itself.

S3 Glacier storage classes ARCHIVE

Very low-cost S3 storage for data you rarely touch.

Think of it as: an off-site storage unit: cheap to keep, but getting things back can take minutes to hours (except Instant Retrieval).

Note: Use Glacier through S3. The old standalone Glacier 'vault' service closed to new customers in December 2025.

Databases

Where your app remembers things



Amazon RDS MANAGED SQL

Managed SQL databases. Backups and patching handled.

Use it when: you want MySQL, PostgreSQL, MariaDB, SQL Server, Oracle or Db2 without babysitting a server.

Amazon Aurora CLOUD-NATIVE SQL

AWS-built database that speaks MySQL or PostgreSQL.

Use it when: you need more speed and resilience than standard RDS. Storage grows automatically; Serverless scales for you.

Amazon DynamoDB NOSQL

Serverless key-value database that stays fast at any scale.

Think of it as: a giant lookup table: give it a key, get the item back in milliseconds. Ideal for baskets, sessions and profiles.

Amazon ElastiCache IN-MEMORY CACHE

Super-fast in-memory cache (Valkey, Redis OSS, Memcached).

Think of it as: keeping your most-used papers on your desk instead of walking to the filing cabinet every time.

Networking & delivery

How users reach your app, quickly and safely



Amazon VPC PRIVATE NETWORK

Your own private, isolated network inside AWS.

Think of it as: your office building: you design the floors (subnets), the doors and who gets in (security groups).

Amazon Route 53 DNS

Turns names like `example.com` into server addresses.

Think of it as: the internet's phone book. It can also register domains and steer traffic away from failures.

Amazon CloudFront CDN

Caches your content close to users around the world.

Think of it as: local shops stocking your goods, so customers don't trek to the central warehouse. Faster pages.

Elastic Load Balancing TRAFFIC SPREADER

Spreads traffic across servers and skips unhealthy ones.

Think of it as: a supermarket queue manager sending each shopper to the next free till.

Amazon API Gateway API FRONT DOOR

Front door for your APIs: routing, security and throttling.

Think of it as: hotel reception: checks who you are, then sends your request to the right room (often a Lambda).

Security & identity

Who can do what, and keeping data safe



AWS IAM PERMISSIONS

Controls who can do what in your AWS account.

Think of it as: office key cards: everyone gets only the rooms they need. Tip: never use the root user day to day.

AWS KMS ENCRYPTION KEYS

Creates and controls the keys that encrypt your data.

Think of it as: a bank vault for your keys, with a record of every use. Built into S3, EBS, RDS and more.

Amazon Cognito APP LOGINS

Sign-up, sign-in and user management for your apps.

Think of it as: a ready-made login system, including 'Sign in with Google or Apple', so you don't build your own.

AWS Secrets Manager PASSWORDS & KEYS

Stores passwords and API keys safely, and can rotate them.

Use it when: you're tempted to paste a database password into your code or config file. Don't!

AWS WAF & AWS Shield WEB PROTECTION

WAF blocks bad web requests; Shield fights DDoS attacks.

Think of it as: nightclub security: WAF checks each guest, Shield holds back the crowd surge. Basic Shield is free and automatic.

Monitoring & operations

Knowing what's happening, and why



Amazon CloudWatch

METRICS & LOGS

Collects metrics and logs, draws dashboards, raises alarms.

Think of it as: your car's dashboard and warning lights. Example: alert me if CPU stays above 80% for 5 minutes.

AWS CloudTrail

AUDIT TRAIL

Records who did what, when and from where in your account.

Think of it as: CCTV for your AWS account. Perfect for audits and 'who deleted that?' moments.

AWS Config

COMPLIANCE

Tracks resource settings over time and checks them against rules.

Think of it as: a building inspector with a history file: 'Is any S3 bucket public? When did that change?'

AWS Systems Manager

SERVER ADMIN

Patch, manage and securely connect to servers at scale.

Use it when: you need to patch 200 servers tonight, or log in securely without opening SSH ports.

Messaging & integration

How the parts of your app talk to each other



Amazon SQS QUEUE

A queue: one part of your app leaves work, another picks it up.

Think of it as: the order rail in a restaurant kitchen: waiters add tickets, chefs take them when ready. Nothing lost in a rush.

Amazon SNS PUB/SUB

Publish one message, deliver it to many subscribers at once.

Think of it as: a group text: one message goes out to email, SMS, mobile push, queues or Lambda functions.

Amazon EventBridge EVENT ROUTER

Routes events like 'order placed' to the right targets by rules.

Think of it as: a smart post room sorting mail to the right department. Its Scheduler also replaces old cron jobs.

AWS Step Functions WORKFLOWS

Runs multi-step workflows with retries and error handling.

Think of it as: a recipe card: do step 1, then step 2; if something fails, retry or take another path. Drawn visually.

AI & machine learning

Adding intelligence, no PhD required



Amazon Bedrock

GENERATIVE AI

Use leading AI models from many providers through one API.

Use it when: building chatbots, document summaries or AI agents without training or hosting models yourself.

Amazon SageMaker AI

CUSTOM ML

Build, train and deploy your own machine-learning models.

Think of it as: a fully equipped workshop for data scientists, for when an off-the-shelf model isn't enough.

Amazon Rekognition

VISION

Analyses images and video: objects, text, faces, unsafe content.

Use it when: auto-tagging photos, moderating user uploads or checking a selfie matches an ID photo.

Amazon Comprehend

TEXT INSIGHTS

Finds meaning in text: sentiment, key phrases, personal data.

Use it when: sorting thousands of customer reviews by mood.

Kiro (from AWS)

AI CODING

AI coding IDE and CLI that turns specs into working code.

Use it when: you want an AI pair programmer. It's AWS's successor to the Amazon Q Developer IDE plugins.

Why no Amazon Q? Q Business closed to new customers in July 2026.

Developer & deployment tools

Building and shipping, the repeatable way



AWS CloudFormation

INFRASTRUCTURE AS CODE

Describe your setup in a template; AWS builds it for you.

Think of it as: flat-pack instructions: the same result every time, and easy to rebuild or tear down.

AWS CDK

IAC IN REAL CODE

Define infrastructure in TypeScript, Python, Java, C# or Go.

Use it when: you'd rather write loops and functions than hundreds of lines of YAML. It generates CloudFormation for you.

AWS CodePipeline

CI/CD

Automates build, test and deploy on every code change.

Think of it as: a factory assembly line from your commit to production, with checks at every station.

Amazon ECR

CONTAINER IMAGES

Private registry to store and scan your container images.

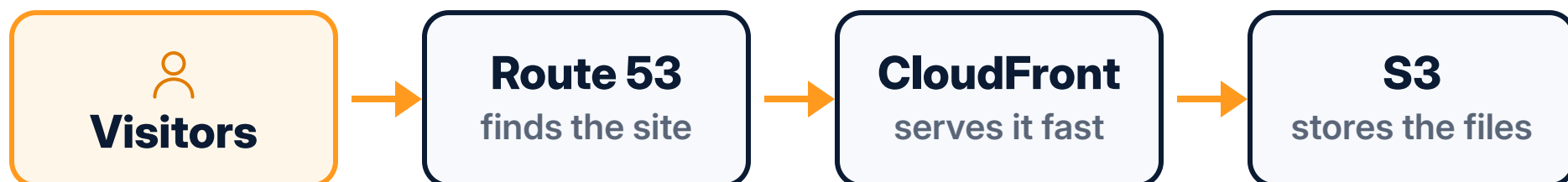
Think of it as: a secure warehouse for your Docker images, ready for ECS, EKS or Lambda to pick up.

Common combos

Three real-world stacks you'll see again and again

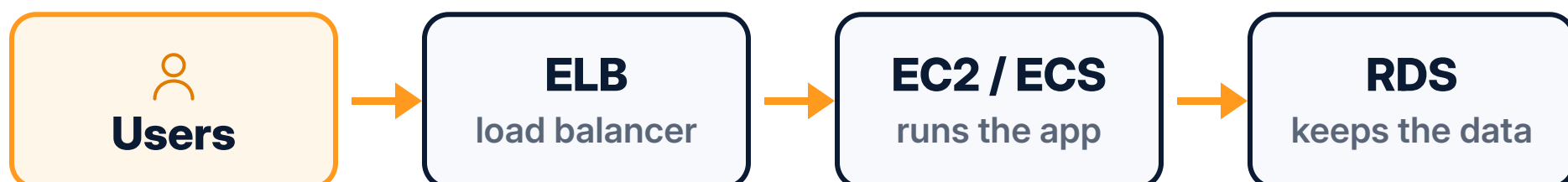


1 Simple website



Good for: portfolios, landing pages, docs. No servers to patch, and very cheap at low traffic.

2 Classic web app



Good for: most business apps and CMS sites. Run servers in 2+ Availability Zones for resilience.

3 Serverless API



Good for: mobile back ends and spiky traffic. Pay per request; scales up, and down to almost nothing.

Add to all three: IAM for permissions, CloudWatch for monitoring.

Found this **useful**?



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About me: web developer from 1994 to 2016, now an application support and tech lead. I explain tech the way I wish someone had explained it to me.

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